UNIVERSITÉ DE GENÈVE

ANALYSE ET TRAITEMENT DE L'INFORMATION
14X026

TP A: Linear Algebra

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Exercise 1. Theory questions

Use the slides and your notes to answer these questions. You do not need to write code. Provide short explanations and examples where possible.

1. What is a vector? What is a vector space? Give an example of a vector in $\mathbb{R}^2$ or $\mathbb{R}^3$.

We can define vector as a mathematical object that has magnitude, which can also be reffered as length, and length.To give an example $\vec{v} = (3,4)$ is a vector in two dimensional space.

Vector Space is a set that contains vectors $(over \mathbb{R})$. Moreover, we are expecting the vector space to satisfy two requirement:

Vector addition: for vector addition when two vectors are added from the same vector space their result should be also in the same vector space $u, v \in V$ then $u + v \in V$

Scalar multiplication: when a vector is mutliplied with a value which is scalar should be in the same vector space with the initial vector. So, if $v \in V$ then there is a value $\alpha \in \mathbb{R}$ when they are multiplied: $\alpha v \in V$

On the other hand, to give an example to the vector can be:

$$\vec{w} = (1, 0, -3) \in \mathbb{R}^3$$

2. What do we mean by the "dimension" D of a dataset?

Dimension is the total number of features used to identify each example(data point) in a dataset. In example, when we consider:

$$X = [x_1, x_2, x_3, ..., x_i] \qquad and \qquad x_i \in \mathbb{R}^D$$

for each data point we can claim that they have D features.

3. Create two small datasets: one scalar dataset of 5 numbers, and one vector dataset of 5 vectors in $\mathbb{R}^3$. Compute their mean and variance. Then, write the formal mathematical definitions of mean and variance.

Example scalar vector dataset is given below:

$$X = \{1, 2, 3, 4, 5\}$$

Average equation:

$$\bar{x} = \frac{1}{5}\sum_{i=1}^{5} x_i = \frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$$

To find the deviations and their squares using the dataset members subtracted by the average to use in population variance:

$$\begin{aligned} 1-3 &= -2 & (-2)^2 &= 4 \\ 2-3 &= -1 & (-1)^2 &= 1 \\ 3-3 &= 0 & 0^2 &= 0 \\ 4-3 &= 1 & 1^2 &= 1 \\ 5-3 &= 2 & 2^2 &= 4 \end{aligned}$$

Population variance

$$\sigma^2 = \frac{1}{5}\sum_{i=1}^{5}(x_i - \bar{x})^2 = \frac{1}{5}(4+1+0+1+4) = \frac{10}{5} = 2$$

Sample vairance that predicts population variance unbiased (Bessel) (1/(N-1)):

$$s^2 = \frac{1}{4}\sum_{i=1}^{5}(x_i - \bar{x})^2 = \frac{10}{4} = 2.5$$

Vector dataset in ($\mathbb{R}^3$) and step by step calculation is given below:

$$\mathbf{x}^{(1)} = (1,0,2),\ \mathbf{x}^{(2)} = (2,1,3),\ \mathbf{x}^{(3)} = (3,2,4),\ \mathbf{x}^{(4)} = (4,3,5),\ \mathbf{x}^{(5)} = (5,4,6)$$

Average vector:

$$\boldsymbol{\mu} = \frac{1}{5}\sum_{i=1}^{5}\mathbf{x}^{(i)} = \left(\frac{1+2+3+4+5}{5},\ \frac{0+1+2+3+4}{5},\ \frac{2+3+4+5+6}{5}\right)$$
$$= (3,\ 2,\ 4)$$

Centered vectors:

$$\begin{aligned}
\mathbf{d}^{(1)} &= \mathbf{x}^{(1)} - \boldsymbol{\mu} = (-2,-2,-2)\\
\mathbf{d}^{(2)} &= \mathbf{x}^{(2)} - \boldsymbol{\mu} = (-1,-1,-1)\\
\mathbf{d}^{(3)} &= \mathbf{x}^{(3)} - \boldsymbol{\mu} = (0,0,0)\\
\mathbf{d}^{(4)} &= \mathbf{x}^{(4)} - \boldsymbol{\mu} = (1,1,1)\\
\mathbf{d}^{(5)} &= \mathbf{x}^{(5)} - \boldsymbol{\mu} = (2,2,2)
\end{aligned}$$

To demonstrate compact D matrix is given below (columns $\mathbf{d}^{(i)}$):

$$D = \begin{bmatrix} -2 & -1 & 0 & 1 & 2\\ -2 & -1 & 0 & 1 & 2\\ -2 & -1 & 0 & 1 & 2 \end{bmatrix}$$

$DD^\top$ (here the calculate matrix's pultiplication with its transpose the one random member of the matrix):

$$(DD^\top)_{11} = (-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2 = 4+1+0+1+4 = 10$$

$$DD^\top = \begin{bmatrix} 10 & 10 & 10\\ 10 & 10 & 10\\ 10 & 10 & 10 \end{bmatrix}$$

Population covariance :

$$\Sigma = \frac{1}{5}DD^\top = \begin{bmatrix} 2 & 2 & 2\\ 2 & 2 & 2\\ 2 & 2 & 2 \end{bmatrix}$$

Sample Covariance(1/(N-1)):

$$S = \frac{1}{4}DD^\top = \begin{bmatrix} 2.5 & 2.5 & 2.5\\ 2.5 & 2.5 & 2.5\\ 2.5 & 2.5 & 2.5 \end{bmatrix}$$

4. Define the scalar product. Compute it for two vectors from your vector dataset above. How does the scalar product allow us to compute the projection of one vector onto another?

Scalar product and projection

Definition: For $\mathbf{u},\mathbf{v} \in \mathbb{R}^D$,

$$\mathbf{u}\cdot\mathbf{v} = \sum_{i=1}^{D} u_i v_i, \qquad \|\mathbf{u}\|^2 = \mathbf{u}\cdot\mathbf{u}.$$

Two vectors from the dataset:

$$\mathbf{a} = \mathbf{x}^{(2)} = (2,1,3), \qquad \mathbf{b} = \mathbf{x}^{(4)} = (4,3,5).$$

Dot product:

$$\mathbf{a}\cdot\mathbf{b} = 2\cdot 4 + 1\cdot 3 + 3\cdot 5 = 8+3+15 = 26.$$

Projection of $\mathbf{a}$ *onto* $\mathbf{b}$*:*

$$\mathrm{proj}_{\mathbf{b}}(\mathbf{a}) = \frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{b}\|^2}\,\mathbf{b}, \qquad \|\mathbf{b}\|^2 = 4^2+3^2+5^2 = 50.$$

$$\text{proj}_{\mathbf{b}}(\mathbf{a}) = \frac{26}{50}(4,3,5) = \frac{13}{25}(4,3,5) = \left(\tfrac{52}{25}, \tfrac{39}{25}, \tfrac{65}{25}\right).$$

Relation with angle:

$$\mathbf{a}\cdot\mathbf{b} = \|\mathbf{a}\|\,\|\mathbf{b}\|\cos\theta, \qquad \|\mathbf{a}\| = \sqrt{14},\ \|\mathbf{b}\| = \sqrt{50}, \qquad \cos\theta = \frac{26}{\sqrt{14}\sqrt{50}}.$$

5. What is a basis of a vector space?

The basis of the vector space is a set that satisfies the conditions below: It should be linearly independent and none of the vectors summation should give any vectors in this vector space. Which means it represents different directions. Also, it spans all of the space using this basis vectors' linear combination

6. Explain the eigenvalue decomposition (EVD) and singular value decomposition (SVD). Why are they important in data science?

Eigenvalue Decomposition (EVD): When the square matrix $A \in \mathbb{R}^{n\times n}$ is considered, that is diagonalizable (it can be written in a simpler way). The equation below should be satisfied:

$$A = V\Lambda V^{-1},$$

where V contains the eigenvectors of A and Λ is a diagonal matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$. So, each eigenpair satisfies

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Singular Value Decomposition (SVD): For any matrix $M \in \mathbb{R}^{m\times n}$,

$$M = U\Sigma V^{\top},$$

Here U is an $m \times m$ orthogonal matrix, V is an $n \times n$ orthogonal matrix, and Σ is an $m \times n$ diagonal matrix containing the singular values such as $\sigma_1, \ldots, \sigma_{\min(m,n)}$. And When we talk about the importance of them in data science we should mention:

- Principal Component Analysis (PCA) is based on SVD/EVD.
- They are also important for noise reduction since they truncate small singular values

7. Give one concrete example of the curse of dimensionality, and one example of the blessing of dimensionality.

Write your answer here

8. If you divide a D-dimensional space into b bins per dimension, and want k samples per bin, how many samples are needed in total?

Write your answer here

9. Why does data become more sparse as D grows? What happens if the number of samples stays fixed?

Write your answer here

10. Compare a hypersphere of radius r with its enclosing hypercube $[-r, r]^D$. As D increases, what happens to their volume ratio? What does this imply for uniform sampling?

Write your answer here

11. In high dimensions, what is the typical angle between two random vectors? How does this angle change as D increases?

Write your answer here

12. What is meant by a “concentration phenomenon"? Give one example where it is useful, and one where it is problematic.

Write your answer here

13. State one concentration inequality (e.g., Markov, Chebyshev, Hoeffding). Explain briefly what it tells us.

Write your answer here

14. What is a PAC (Probably Approximately Correct) analysis in machine learning?

Write your answer here

Exercise 2. Matrix

1. Find a quadratic polynomial, say $f(x) = ax^2 + bx + c$, such that

$$f(1) = -5, \quad f(2) = 1, \quad f(3) = 11.$$

We know the informations below and for random quadratic polynomial we should use the standart $ax^2 + bx + c$ to get the real equation.

$$f(x) = ax^2 + bx + c, \qquad f(1) = -5,\ f(2) = 1,\ f(3) = 11$$

Here the values which were given has been written into equation to find the real equation.

$$\begin{aligned} f(1) &= a + b + c = -5 \\ f(2) &= 4a + 2b + c = 1 \\ f(3) &= 9a + 3b + c = 11 \end{aligned}$$

Then we are converting these equation to find the exact equation and we are reducing the matrix (the gauss-jordan elimination has applied to get the row reduced echelon form).

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & -5 \\ 4 & 2 & 1 & 1 \\ 9 & 3 & 1 & 11 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 4R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & -5 \\ 0 & -2 & -3 & 21 \\ 9 & 3 & 1 & 11 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow R_3 - 9R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & -5 \\ 0 & -2 & -3 & 21 \\ 0 & -6 & -8 & 56 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow -\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & -5 \\ 0 & 1 & \frac{3}{2} & -\frac{21}{2} \\ 0 & -6 & -8 & 56 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow R_3 + 6R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & -5 \\ 0 & 1 & \frac{3}{2} & -\frac{21}{2} \\ 0 & 0 & 1 & -7 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow R_2 - \frac{3}{2}R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & -5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -7 \end{array}\right]$$

$$\xrightarrow{R_1 \leftarrow R_1 - R_3} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -7 \end{array}\right]$$

$$\xrightarrow{R_1 \leftarrow R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -7 \end{array}\right]$$

Here we got the values below from the matrix:

$$a = 2, \quad b = 0, \quad c = -7$$

and we can write the equation like:

$$f(x) = 2x^2 - 7$$

To make it accurate we can try the values which has been given in the question like below:

$$f(1) = 2 - 7 = -5$$
$$f(2) = 8 - 7 = 1$$
$$f(3) = 18 - 7 = 11$$

2. Let a, b be some fixed parameters. Solve the system of linear equations

$$\begin{cases} x + ay = 2 \\ bx + 2y = 3 \end{cases}$$

Here again to solve this equation we should apply gauss-jordan elimination like below:

$$\left[\begin{array}{cc|c} 1 & a & 2 \\ b & 2 & 3 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - bR_1} \left[\begin{array}{cc|c} 1 & a & 2 \\ 0 & 2-ab & 3-2b \end{array}\right] \xrightarrow{R_2 \leftarrow \frac{1}{2-ab} R_2} \left[\begin{array}{cc|c} 1 & a & 2 \\ 0 & 1 & \frac{3-2b}{2-ab} \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 - aR_2} \left[\begin{array}{cc|c} 1 & 0 & \frac{4-3a}{2-ab} \\ 0 & 1 & \frac{3-2b}{2-ab} \end{array}\right]$$

$$\boxed{x = \frac{4-3a}{2-ab}}, \qquad \boxed{y = \frac{3-2b}{2-ab}} \qquad (ab \neq 2)$$

$$ab = 2 \Rightarrow \begin{cases} bx + 2y = 2b \\ bx + 2y = 3 \end{cases}$$

$$2b \neq 3 \Rightarrow \varnothing$$

$$2b = 3 \Rightarrow b = \tfrac{3}{2},\ a = \tfrac{4}{3},\ x + \tfrac{4}{3}y = 2 \Rightarrow (x, y) = \left(2 - \tfrac{4}{3}t,\ t\right),\ t \in \mathbb{R}.$$

3. Find the inverse of the following matrix, if it exists:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$$

To find the inverse of this matrix we are finding the determinant to test this matrix that whether it has an inverse or not. It does not give 0 value so that means it has an inverse.

$$\det(A) = 1\begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} - 2\begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} + 3\begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix} = -24 + 40 - 15 = 1$$

Here the matrix was reduced using gauss jordan elimination to find the inverse.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 5 & 6 & 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - 5R_1} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & -4 & -15 & -5 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow R_3 + 4R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & -5 & 1 & -2 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow R_2 - 4R_3,\ R_1 \leftarrow R_1 + 5R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -24 & 18 & 5 \\ 0 & 1 & 0 & 20 & -15 & -4 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array}\right]$$

After the reducing process finished we got the matrix below:

$$A^{-1} = \begin{pmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{pmatrix}$$

4. Solve the following matrix equation for X:

$$AX = B, \quad \text{where} \quad A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 7 & 8 \\ 9 & 10 \end{pmatrix}.$$

To find the inverse of a we should find the determinant:

$$\det(A) = \begin{vmatrix} 1 & 3 \\ 2 & 5 \end{vmatrix} = 5 - 6 = -1$$

An then the formula below was utilized to find the inverse of the matrix:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \det(A) = ad - bc \neq 0, \qquad A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Placing the values to the formula:

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} 5 & -3 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix}$$

And to find the X the equation $A^{-1}B$ should be solved

$$X = A^{-1}B = \begin{pmatrix} -5 & 3 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 7 & 8 \\ 9 & 10 \end{pmatrix} = \begin{pmatrix} -5 \cdot 7 + 3 \cdot 9 & -5 \cdot 8 + 3 \cdot 10 \\ 2 \cdot 7 - 1 \cdot 9 & 2 \cdot 8 - 1 \cdot 10 \end{pmatrix} = \begin{pmatrix} -8 & -10 \\ 5 & 6 \end{pmatrix}$$

$$\boxed{X = \begin{pmatrix} -8 & -10 \\ 5 & 6 \end{pmatrix}}$$

Exercise 3. The importance of the mathematical concept behind a code

1. Download, open and run several times the PYTHON file "some_script.py".

   Explain the function `def project_on_first(u, v)` in geometrical terms and then explain the results of this code with clear mathematical concepts (you have to talk about projections and scalar product). *It is very important to understand the mathematical concepts behind a program!!*

Write your solution here When we consider the compilation of the program we can observe that the code always gives the result as below:

Result = 0.0

Result = 0.0

That was caused by vectors u and w's ortogonality. To explain better the code should be investigated as below:

(a) initially the code first sets the working space as two-dimensional plane by using the line below:

- N=2

(b) then the code selects random vectors for the u and v:

```
u = np.zeros(N)
u[np.random.randint(N)] = 1
```

Here with these lines u has been set as a canonical vector which means it can be only:

$$[1,0] \quad \text{(in the x axis direction)}$$

$$[0,1] \quad \text{(in the y axis direction)}$$

For the v the code directly sets the v as a random vector using the line below:

$$v = \text{np.random.randn}(N) \quad \text{(i.e. } [-2,5],[4,6],\ldots)$$

(c) Using these vectors there are some methods that has been defined:

i. Scalar product function

```
def scalar_product(u, v):
    return np.dot(u, v)
```

this function calculates the scalar multiplication of the two vectors.

ii. Project on first function:

```
def project_on_first(u, v):
    return np.dot(u,v) / np.dot(u,u) * u
```

This function has been utilized to show the equation of projection of the v over u
this can be explained as mathematically we are searching a vector that is over u but it thould be the closest to the v vector. Here if we create this vector as "ku" the ecuation can be demonstrated as:

$$u \cdot (v - ku) = 0 \;\Rightarrow\; u \cdot v - ku \cdot u = 0 \;\Rightarrow\; a = \frac{u \cdot u}{u \cdot v}$$

Therefore we can say that

$$proj_u(v) = ku$$

iii. Deviate vector function:

```
def deviate_vector(u, v):
    return v - project_on_first(u, v)
```

This function deletes the part parallel to u from v, leaving only the part perpendicular to u. In this way, w will be equal to this value.
And using this method when the u and w was multiplied in the code as below:

```
w = deviate_vector(u, v)
print('Result = %.1f' % scalar_product(u, w))
```

the code always gives the 0 value which comes from the equation series below:

$$u \cdot w = u \cdot (v - proj_u(v))$$

here the value was distributed:

$$u \cdot w = u \cdot v - u \cdot proj_u(v)$$

using this equation we can write projection as:

$$proj_u(v) = \left(\frac{u \cdot u}{u \cdot v}\right) u$$

then:

$$u \cdot proj_u(v) = u \cdot \left(\frac{u \cdot v}{u \cdot u} u\right)$$

and if we take out the scalar constant:

$$= \frac{u \cdot v}{u \cdot u}(u \cdot u) = u \cdot v$$

Finally if we write this equation in the stage below:

$$u \cdot w = u \cdot v - u \cdot proj_u(v)$$

$$u \cdot w = u \cdot v - u \cdot v = 0$$

which represents that the w will be always perpendicular to the u.
With numbers it can be demonstrated as an example below:
Lets say:

- u = (1,0)
- v = (3,4)

$$proj_u(v) = (3,0)$$

$$w = v - proj_u(v) = (0,4)$$

and if we test these values:

$$u \cdot w = (1,0) \cdot (0,4) = 0$$

So, even if u and v are different in each run, the result is:
The projection always produces a vector "parallel to u."
The remaining w automatically remains "perpendicular" to u.
And according to the demonstrations that has been done the result is always perpendicular.

(d) Consider $u = (u_i)_{1 \le i \le 7}$ and $v = (v_i)_{1 \le i \le 7}$, two vectors of length 7. How would you clearly write the following part of the script using some mathematical notation?

```
# import numpy library as np in order to work on tensors
import numpy as np
# create 2 random vectors of length 7
u = np.random.randn(7)
v = np.random.randn(7)
# perform some computations
r = 0
for ui, vi in zip(u, v) :
    r += ui * vi
```

Name the mathematics behind and rewrite the last 3 lines of code in one.

Write your solution here Here in this code we are actually doing a dot product in the lines below:

```
r = 0
for ui, vi in zip(u, v):
    r += ui * vi
```

Here we are doing for all i a ui and vi multiplication and we are summing up all these multiplications. This is called scalar product in mathematics. In example for two

$$u = (u_1, \ldots, u_7), \quad v = (v_1, \ldots, v_7)$$

vectors:

$$u \cdot v = \sum_{i=1}^{7} u_i v_i$$

and if we have to write this clearly we do not need any for loop instead we can use the NumPy library function below:

```
r = np.dot(u, v)
```

(e) Complete the code to construct a vector v orthogonal to the vector u and with the same norm as u. Comment each line of your code.

Write your solution here

```
import numpy as np

# Here for the u vector we are creating a vector which has a random length
u = np.random.randn(7)

# But for this one we are creating an another vector which is random also
v0 = np.random.randn(7)

# proj_u(v0) = (u·v0 / u·u) * u
# here the in the code the v vector was created to be perpendicular to u
# to create this vector the sense of the code line used below:
# w = v0 - proj_u(v0) also we know that:
# proj_u(v0) = (u·v0 / u·u) * u
v = v0 - (np.dot(u, v0) / np.dot(u, u)) * u

# And now the code satisfied the perpendicularity of the vectors but
# they should also have same length so we should satisfy:
# ||v|| = ||u||
# np.linalg.norm(u) represents the ||u||
# so we should rearrange the norms as below code by using the ratio
# between ||u||/||v|| multiplied with v to get the same length with u.
v = v * (np.linalg.norm(u) / np.linalg.norm(v))

# here the code was checked using print functions:
print("u · v = 0", np.allclose(np.dot(u,v), 0)) # all close function checks whether
#the dot product equals to zero or not that demonstrates perpendicularity.
print("||u|| =", np.linalg.norm(u)) # length of u
print("||v|| =", np.linalg.norm(v)) # length of v (the results should be equal to get the prop
```

When I run this code 2 times I got the outputs below:

```
u · v = 0 True
||u|| = 1.7881886907208657
||v|| = 1.7881886907208657

u · v = 0 True
||u|| = 2.8554131029671903
||v|| = 2.8554131029671903
```

which shows that it is working properly and gives the proper values.

(f) Given two vectors u and v in $\mathbb{R}^n$, write Python code to compute the cosine of the angle between them. Explain the mathematical concept behind this computation, and show how you would write it in one equation.

Write your solution here To find the cosine of the angle between u and v we should think the equation below:

i. The u and v can be represented using norms and the angles as the (x,y) coordinates such as:

$$u = (\|u\| \cos\alpha, \|u\| \sin\alpha)$$

$$v = (\|v\| \cos\beta, \|v\| \sin\beta)$$

ii. when we write the dot product as below:

$$u \cdot v = u_1 v_1 + u_2 v_2$$

and write the coordinates inside of the equation we get:

$$u \cdot v = \|u\| \cos\alpha \cdot \|v\| \cos\beta + \|u\| \sin\alpha \cdot \|v\| \sin\beta$$

then when we put it in parentheses of $\|u\|\|v\|$ we can get:

$$u \cdot v = \|u\|\|v\|(\cos\alpha\cos\beta + \sin\alpha\sin\beta)$$

and by the trigonometrical rule below:

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

so we can write the "$u \cdot v = \|u\|\|v\|(\cos\alpha\cos\beta + \sin\alpha\sin\beta)$" equation as:

$$u \cdot v = \|u\|\|v\| \cos(\alpha - \beta)$$

α is the angle that u makes with the x-axis. on the other hand, β is the angle made by v with the x-axis therefore when to get the angle between them we should subtract them as below:

$$\theta = \beta - \alpha$$

this equation shows that theta equals the angle between them so we can write the equation "$u \cdot v = \|u\|\|v\| \cos(\alpha - \beta)$" as below:

$$u \cdot v = \|u\|\|v\| \cos\theta$$

With this equation we can compute the cosine angle between them as:

$$\cos\theta = \frac{u \cdot v}{\|u\|\|v\|}$$

and to write this equation in one line we can write this line of the code as:

```
cos_theta = np.dot(u, v) / (np.linalg.norm(u) * np.linalg.norm(v))
```

- np.dot(u, v) : represents $u \cdot v$
- np.linalg.norm(u) : represents $\|u\|$
- np.linalg.norm(v) : represents $\|v\|$

Exercise 4. Computing Eigenvalues, Eigenvectors, and Determinants

(a) Find the determinant, eigenvectors, and eigenvalues of the matrix

$$\begin{bmatrix} 5 & 6 & 3 \\ -1 & 0 & 1 \\ 1 & 2 & -1 \end{bmatrix}$$

Compare your results with those of the computer.

Write your solution here **a)** For the determinant the equation should be calculated using cofactors below:

$$\det(A) = \begin{vmatrix} 5 & 6 & 3 \\ -1 & 0 & 1 \\ 1 & 2 & -1 \end{vmatrix}.$$

Sub determinants should be calculated as below:

$$\begin{vmatrix} 0 & 1 \\ 2 & -1 \end{vmatrix} = 0 \cdot (-1) - 1 \cdot 2 = -2 \;\Rightarrow\; 5 \cdot (-2) = -10$$

$$\begin{vmatrix} -1 & 1 \\ 1 & -1 \end{vmatrix} = (-1)(-1) - (1)(1) = 1 - 1 = 0 \;\Rightarrow\; -6 \cdot 0 = 0$$

$$\begin{vmatrix} -1 & 0 \\ 1 & 2 \end{vmatrix} = (-1)(2) - (0)(1) = -2 \;\Rightarrow\; 3 \cdot (-2) = -6$$

And finally when we sum up the determinants we will get the result as:

$$\det(A) = -10 + 0 - 6 = -16$$

b) For the eigenvectors of the matrix we should do the calculation below:
If we consider a matrix A, it takes a vector x and generally it changes the direction or it may make it longer or shorter and even it can rotate the vector.
For example,

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

matrix can rotate a vector 90 degrees.
And there are some special vectors called as eigenvectors which never rotate when the matrix is applied on them. Only the length (stretched/shrunk) of the vector changes. We call the stretching and shrinking factor as eigenvalue.
Therefore, when we use this information and consider the equation below:

$$Ax = \lambda x, \quad (x \neq 0)$$

The matrix makes the length of the vector λ times bigger or shorter. And when we collect two sides to one side we can get the equation below:

$$(A - \lambda I)x = 0$$

This shows the eigenvector x is part of a homogeneous linear system solution.
And if we consider eigenvectors, the vector x should not be equal to 0. For this situation, the system should not have only the trivial solution and the matrix should not be invertible. Therefore, the equation below should be valid for the matrix:

$$\det(A - \lambda I) = 0$$

The λ values that come from this equation are the eigenvalues. And the results will be a polynomial and we will call this polynomial a characteristic polynomial since it contains (eigenvalues, trace, determinant):

$$p(\lambda) = \det(A - \lambda I)$$

Here with these steps we are creating the characteristic polynomial using the identity matrix I multiplied with λ and subtracted from the matrix A.

The below steps have been carried out to find the eigenvalues.

$$A = \begin{pmatrix} 5 & 6 & 3 \\ -1 & 0 & 1 \\ 1 & 2 & -1 \end{pmatrix}$$

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\lambda I = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$$

$$\lambda I - A = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} - \begin{pmatrix} 5 & 6 & 3 \\ -1 & 0 & 1 \\ 1 & 2 & -1 \end{pmatrix}$$

$$\lambda I - A = \begin{pmatrix} \lambda - 5 & -6 & -3 \\ 1 & \lambda & -1 \\ -1 & -2 & \lambda + 1 \end{pmatrix}$$

Here with this steps we are creating characteristic polynomial using I(identity matrix) multiplied with λ and subtracted from the A matrix.

$$p(\lambda) = \det(\lambda I - A)$$

we took the determinant of the above equation and got the equation below:

$$p(\lambda) = (\lambda - 5) \begin{vmatrix} \lambda & -1 \\ -2 & \lambda + 1 \end{vmatrix} - (-6) \begin{vmatrix} 1 & -1 \\ -1 & \lambda + 1 \end{vmatrix} + (-3) \begin{vmatrix} 1 & \lambda \\ -1 & -2 \end{vmatrix}$$

then,

$$= (\lambda - 5)\,[\lambda(\lambda + 1) - (-1)(-2)] + 6\,[1(\lambda + 1) - (-1)(-1)] + (-3)\,[1(-2) - \lambda(-1)]$$

here below the expression was expanded,

$$= (\lambda - 5)(\lambda^2 + \lambda - 2) + 6(\lambda + 1 - 1) + (-3)(-2 + \lambda)$$

$$= (\lambda - 5)(\lambda^2 + \lambda - 2) + 6\lambda + (-3\lambda + 6)$$

$$= \lambda^3 + \lambda^2 - 2\lambda - 5\lambda^2 - 5\lambda + 10 + 6\lambda - 3\lambda + 6$$

$$= \lambda^3 - 4\lambda^2 - 7\lambda + 10 + 6\lambda - 3\lambda + 6$$

$$p(\lambda) = \lambda^3 - 4\lambda^2 - 4\lambda + 16$$

$$p(\lambda) = (\lambda - 4)(\lambda^2 - 4)$$

$$p(\lambda) = (\lambda - 4)(\lambda - 2)(\lambda + 2)$$

finally the eigenvalues has been found like below:

$$\lambda = 4, \quad \lambda = 2, \quad \lambda = -2$$

c) to find the eigenvectors the eigenvalues (4, 2, -2) will be used to get the equations as like below:

$$A = \begin{pmatrix} 5 & 6 & 3 \\ -1 & 0 & 1 \\ 1 & 2 & -1 \end{pmatrix}$$

Here we are aiming to find null space of the $A - 4I$ below. We are trying to get the row echelon form of the matrix by reducing it.

$$\text{For } \lambda = 4: \quad A - 4I = \begin{pmatrix} 1 & 6 & 3 \\ -1 & -4 & 1 \\ 1 & 2 & -5 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} 1 & 6 & 3 \\ 0 & 2 & 4 \\ 1 & 2 & -5 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 6 & 3 \\ 0 & 2 & 4 \\ 0 & -4 & -8 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftarrow \frac{1}{2} R_2} \begin{pmatrix} 1 & 6 & 3 \\ 0 & 1 & 2 \\ 0 & -4 & -8 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + 4R_2} \begin{pmatrix} 1 & 6 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 6R_2} \begin{pmatrix} 1 & 0 & -9 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

here above by reducing the matrix we got the equations that will give us the first eigenvector. and let z be the free variable and equal to t so: $(x = 9t, y = -2t, z = t)$

$$\Rightarrow \begin{cases} x - 9z = 0 \\ y + 2z = 0 \end{cases} \quad \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 9 \\ -2 \\ 1 \end{pmatrix} t, \ t \in \mathbb{R}, \quad v_1 = \begin{pmatrix} 9 \\ -2 \\ 1 \end{pmatrix}.$$

$$\text{For } \lambda = 2: \quad A - 2I = \begin{pmatrix} 3 & 6 & 3 \\ -1 & -2 & 1 \\ 1 & 2 & -3 \end{pmatrix} \xrightarrow{\text{swap the} R_1 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & -3 \\ -1 & -2 & 1 \\ 3 & 6 & 3 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_1, \, R_3 \leftarrow R_3 - 3R_1} \begin{pmatrix} 1 & 2 & -3 \\ 0 & 0 & -2 \\ 0 & 0 & 12 \end{pmatrix}$$

When we have completed the same equation for $(\lambda = 2)$

$$\xrightarrow{R_2 \leftarrow -\frac{1}{2} R_2} \begin{pmatrix} 1 & 2 & -3 \\ 0 & 0 & 1 \\ 0 & 0 & 12 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 12R_2} \begin{pmatrix} 1 & 2 & -3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + 3R_2} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

We got the equation below from the row echelon form and we can call $z = 2t$ and the eigenvector will be found

$$\Rightarrow \begin{cases} x + 2y = 0 \\ z = 0 \end{cases} \quad \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} t, \ t \in \mathbb{R}, \quad v_2 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}.$$

$$\text{For } \lambda = -2: \quad A + 2I = \begin{pmatrix} 7 & 6 & 3 \\ -1 & 2 & 1 \\ 1 & 2 & 1 \end{pmatrix} \xrightarrow{\text{swap the } R_1 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 1 \\ -1 & 2 & 1 \\ 7 & 6 & 3 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_1, \, R_3 \leftarrow R_3 - 7R_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 4 & 2 \\ 0 & -8 & -4 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 + 2R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 4 & 2 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 \leftarrow \frac{1}{4} R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}$$

For the $(\lambda = -2)$ we got the equations below thanks to row echelon form of the matrix. When we call $y = t$ we will be able to find eigenvector.

$$\Rightarrow \begin{cases} x = 0 \\ y + \frac{1}{2} z = 0 \end{cases} \quad \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} t, \ t \in \mathbb{R}, \quad v_3 = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}.$$

And to compare the values with those of computer we should use the code below:

```
import sympy as sp

A = sp.Matrix([[5,6,3],
               [-1,0,1],
               [1,2,-1]])

print(A.det())
print(A.eigenvals())
print(A.eigenvects())
```

and with this code we will get the values using symbolic mathematic library (sympy) the code gave the outputs below:

```
-16
{4: 1, 2: 1, -2: 1}
[(-2, 1, [Matrix([
[   0],
[-1/2],
[   1]])]),
 (2, 1, [Matrix([
[-2],
[ 1],
[ 0]])]),
 (4, 1, [Matrix([
[ 9],
[-2],
[ 1]])])]
```

The results are the same which shows the accuracy of the computations above. On the other hand, for the λ = -2 the code gives the values divided by 2 for my equation hence this shows that the results are in the same eigenvector space.

(b) The covariance matrix for n samples $x_1, \ldots, x_n$, each represented by a $d \times 1$ column vector, is given by

$$C = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \mu)(x_i - \mu)^T,$$

where C is a $d \times d$ matrix and $\mu = \sum_{i=1}^{n} x_i / n$ is the *sample mean.* Prove that C is always positive semidefinite. (Note: A symmetric matrix C of size $d \times d$ is *positive semidefinite* if $v^T C v \geq 0$ for every $d \times 1$ vector v.)

(c) In this portion of the exercise, we will calculate the eigenvalues of the covariance matrices of six data sets listed as follows:

filename	n	d	description
`tp1_artificialdata[1-3]`	1024	100	Artificial data generated from various auto-regression (AR-1) models
`tp1_artificialdata4`	1024	100	Random Gaussian data
`tp1_freyfaces`	1965	560	Facial images of a man named Brendan
`tp1_digit2`	5958	784	Hand-written images of "2"

To access each data set, download and unzip `data for python.zip`. Each file contains a $n \times d$ data matrix with rows representing n different samples in $\mathbb{R}^d$ For example, `tp1_artificialdata1` contains a data matrix of size 1024×100 (1024 samples, 100 features). Once each dataset has been imported into `Python`, complete the following tasks:

(a) Compute the covariance matrix of each dataset;
(b) Compute the eigenvalues for each covariance matrix;
(c) Compute the determinant of each covariance matrix;
(d) Compute the product of the eigenvalues for each covariance matrix;
(e) Display the eigenspectrum for each covariance matrix in a 2D plot, where the x-axis shows the rank of the eigenvalues, ranging from 1 (the largest eigenvalue) to 100 (the 100-th largest eigenvalue), and the y-axis shows the corresponding eigenvalue.

Describe the relationship between the product of the eigenvalues and the determinant of each covariance matrix. In addition, describe the observations you made regarding the plot of the spectrum of eigenvalues of each covariance matrix. How can you explain the disparities between datasets?

After I execute the python file (ex4.3.pdf) I created a table to make the code give the expected outputs:

covariance_eigenspectra_summary

dataset	n	d	determinant	prod_eigenvalues	prod_equals_det	min_eigenvalue	max_eigenvalue	top_k	spectrum_path
tp1_artificialdata1.npz	1024	100	6.280251291462 37e-39	6.280251291462103e-39	TRUE	0.0978201022491097	7.435351771370377	100	/Users/ilknurdede/Desktop/HWDS1/data for python/results/tp1_artificialdata1_npz_spectrum.png
tp1_artificialdata2.npz	1024	100	2.0261451149199686e-15	2.0261451149199544e-15	TRUE	0.2158144375552696	3.466479461738479	100	/Users/ilknurdede/Desktop/HWDS1/data for python/results/tp1_artificialdata2_npz_spectrum.png
tp1_artificialdata3.npz	1024	100	1.0255780906233115e-05	1.0255780906232788e-05	TRUE	0.37344865147778394	2.1533505440009435	100	/Users/ilknurdede/Desktop/HWDS1/data for python/results/tp1_artificialdata3_npz_spectrum.png
tp1_artificialdata4.npz	1024	100	0.004520320412163924	0.004520320412163851	TRUE	0.47425965983685275	1.7219728644107384	100	/Users/ilknurdede/Desktop/HWDS1/data for python/results/tp1_artificialdata4_npz_spectrum.png
tp1_freyfaces.npz	1965	560	0.0	0.0	TRUE	4.4970226563251 94e-06	1.2858270703135224	100	/Users/ilknurdede/Desktop/HWDS1/data for python/results/tp1_freyfaces_npz_spectrum.png
tp1_digit2.npz	5958	784	0.0	0.0	TRUE	-4.723172476223134e-16	6.104346586238362	100	/Users/ilknurdede/Desktop/HWDS1/data for python/results/tp1_digit2_npz_spectrum.png

Figure 1: Table for results

And for the spectrum paths the below graphs are visible that makes the results can be more accurately considered:

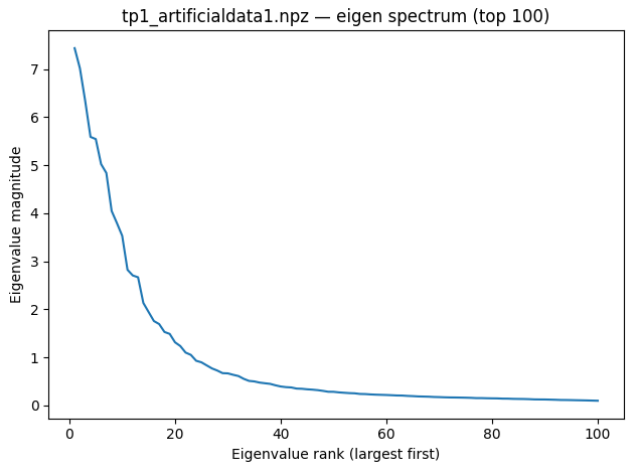

Figure 2

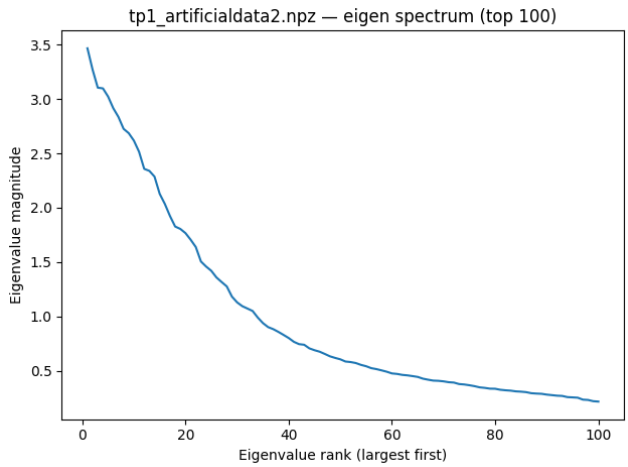

Figure 3

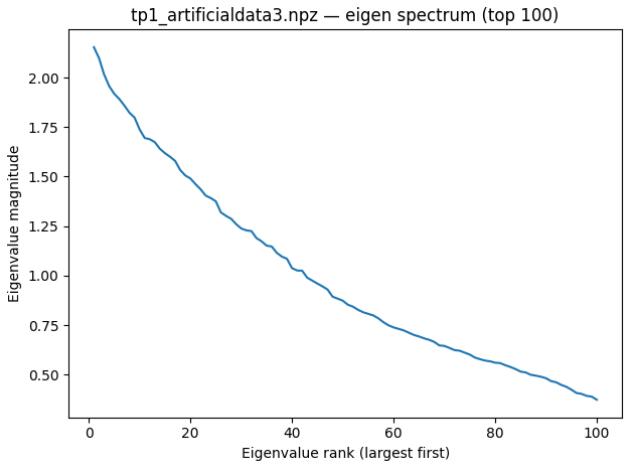

Figure 4

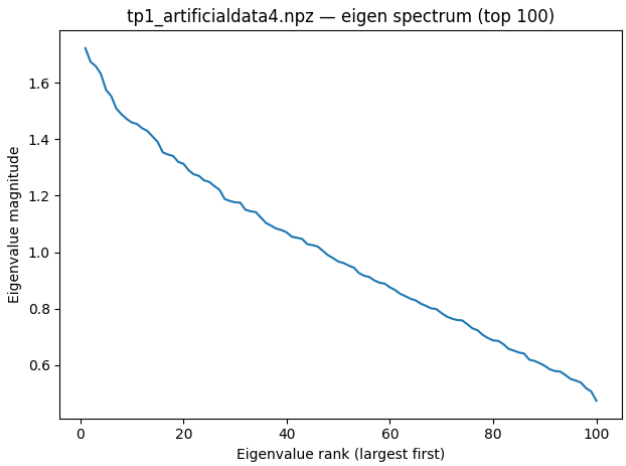

Figure 5

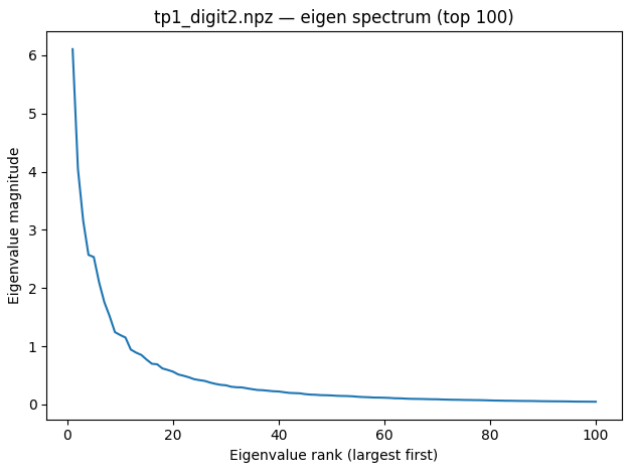

Figure 6

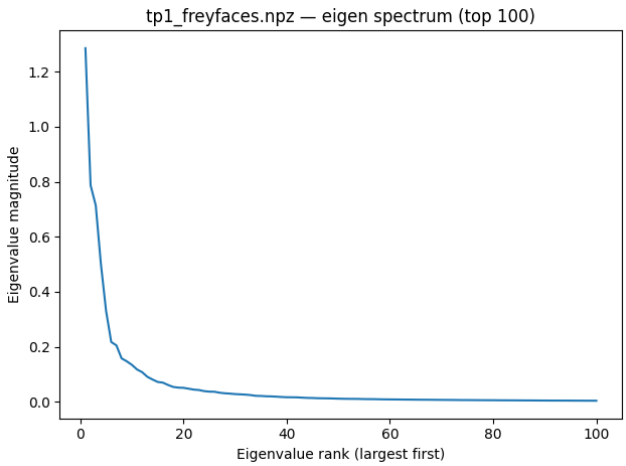
tp1_freyfaces.npz — eigen spectrum (top 100)
Eigenvalue magnitude
Eigenvalue rank (largest first)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0
20
40
60
80
100

Figure 7

Based on previous mathematical experience, we know that the determinant of a covariance matrix is equal to the product of its eigenvalues. The ex4.3.py file I wrote also proves the validity of this claim by returning True for each data set in the productmatchesdet column of the CSV. However, the determinant is 0 for the Freyfaces and Digit2 data sets because the covariance matrix is singular.

According to the graphs result for different data sets we can observe that:

Artificial Data 1: The first few eigenvalues are very large, then there is a rapid decline. This data actually has a few dominant dimensions, with the rest having low variance. PCA dimensionality reduction would be very efficient here.

Artificial Data 2: The first eigenvalues are again somewhat dominant, but the decline is slower. Therefore, this result shows that the data contains meaningful variance across more dimensions.

Artificial Data 3: The eigenvalues form a more evenly decreasing line. There is no clear 'knee' point, meaning the variance is spread across many dimensions.

Artificial Data 4: There is an almost linear decline. The variance is distributed even more evenly.

Digit2: The first few eigenvalues are very large, and they are followed by a long tail. Hence, this can be explained by the presence of highly correlated features in image data . With PCA, a small number of components can represent a large portion of the data.

Freyfaces: There is a very sharp decline, with the first few eigenvalues being several times larger. Face images typically have a low-dimensional structure , which is consistent with the shape in the graph.

When we consider whole of these informations and graphs we can make the comments below: Artificial data: Since they are generated with different distributions, we can observe that the spectrum shapes change in a controlled manner. Some have a few dominant directions, while in others the variance is more evenly distributed. Real image data (Freyfaces, Digit2): These data are very high-dimensional but contain a large amount of redundancy. We can observe that there are strong correlations between pixels. Therefore, it is clear that the covariance matrix is not fully ranked, and the determinant being zero stems from this reason.

So, The determinant and eigenvector product equality can be observed uniformly across all sets. However, in singular matrices, this product is zero.

Eigenspectrum graphs beautifully illustrate the structure of the data. Consequently, we can observe that some data sets are low-dimensional, while others are high-dimensional but redundant.

The differences in artificial data stem from the way they are generated. The differences in real data sets, on the other hand, arise directly from the natural data structure and the high dimensionality, where 'a few eigenvalues are very large but the others are small' and correlations are highly repetitive and very strong.

1. Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. Show that if A has an eigenvalue $\lambda = 0$, then the matrix A is singular. Furthermore, prove that the determinant of A must be zero if any eigenvalue is zero.

Assumptions: $A \in \mathbb{R}^{n\times n}$, $\lambda = 0$ is an eigenvalue of A.

By definition of eigenvalue: $\exists\, v \in \mathbb{R}^n \setminus \{0\}$ such that $Av = \lambda v$.

$\lambda = 0 \Rightarrow Av = 0$.

$v \neq 0$ and $Av = 0 \Rightarrow v \in \ker(A)$ with $v \neq 0 \Rightarrow \ker(A) \neq \{0\}$.

Invertible Matrix Theorem: $\ker(A) = \{0\} \iff A$ is invertible.

Contrapositive: $\ker(A) \neq \{0\} \Rightarrow A$ is not invertible $\Rightarrow A$ is singular.

Determinant via invertibility: A invertible $\iff \det(A) \neq 0$.

Contrapositive: A singular $\Rightarrow \det(A) = 0$.

Direct determinant proof (characteristic polynomial):

$p_A(\lambda) := \det(A - \lambda I)$.

$\lambda = 0$ eigenvalue $\Rightarrow p_A(0) = 0$.

$p_A(0) = \det(A - 0 \cdot I) = \det(A) \Rightarrow \det(A) = 0$.

General statement: $\det(A) = \prod_{i=1}^{n} \lambda_i$ (eigenvalues counted with algebraic multiplicity).

$\exists\, k$ with $\lambda_k = 0 \Rightarrow \prod_{i=1}^{n} \lambda_i = 0 \Rightarrow \det(A) = 0$.

Exercise 5. Computing Projection Onto a Line

There are given a line $\alpha : 3x + 4y = -6$ and a point A with the coordinates (-1, 3).

1. Find a distance from the point A to the line α using linear algebra (not coordinate method).
2. Explain your code using a sketch and some mathematics (there should be a scalar product somewhere).

We initially know the informations below:

$$\text{here given } \alpha : \ 3x + 4y = -6, \quad A = (-1, 3).$$

$$n = \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \qquad \|n\| = \sqrt{3^2 + 4^2} = 5.$$

Here we are defining the normal vector using the 3 from the 3x and 4 from the 4y. From this vector we are calculating the norm of n which is 5

$$\text{Choose } p_0 \in \alpha : \ p_0 = (-2, 0) \ \text{ since } \ 3(-2) + 4 \cdot 0 = -6.$$

Then to get the random coordinate over the line p_0 we are writing $y = 0$ into the y and we can get the coordinate $(-2, 0)$.

$$A - p_0 = \begin{pmatrix} -1 \\ 3 \end{pmatrix} - \begin{pmatrix} -2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

To find the general formula of the distance we should find $A - p_0$ in the formula

$$\text{dist}(A, \alpha) = \frac{|(A - p_0) \cdot n|}{\|n\|},$$

to find the distance form A to the line α.

$$(A - p_0) \cdot n = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 4 \end{pmatrix} = 1 \cdot 3 + 3 \cdot 4 = 15.$$

Here the $A - p_0$ was multiplied with n.

$$\text{dist}(A, \alpha) = \frac{|(A - p_0) \cdot n|}{\|n\|} = \frac{15}{5} = 3.$$

And the result has been found as 3. Moreover to create the sketch the code ex5.py has been written and the skecth that explains the result has been demonstrated below:

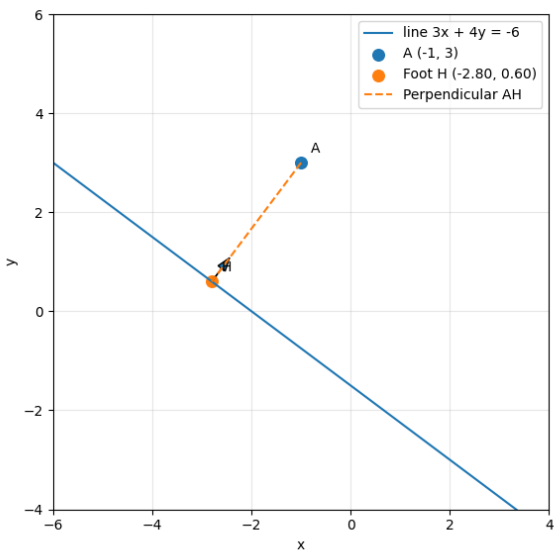

Figure 8